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Urbanization of energy poverty? The case of Mozambique

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ABSTRACT

What is the impact of the unprecedented scale and pace of urbanization in the global South on the evolution of energy poverty incidence? Remarkably enough, urban population growth and energy poverty are mainly studied separately from each other in different strands of the literature. We argue that there is much to be gained in studying these phenomena in their interdependence. To illustrate this, we analyze for the case of Mozambique the relationship between spatial variation in population density and three commonly used indicators of energy poverty over time. We think that the case of Mozambique exemplifies the future development of Africa's energy poverty situation in many respects. Our analysis makes use of two newly constructed datasets with household level information that originate from national household welfare surveys as well as an own survey for Mozambique's capital city Maputo. The data show that over time, and especially in urban areas, energy poverty is decreasing in terms of consumption quantities and access to modern energy fuels, but increasing in terms of energy expenditure shares. Also, we show that by largely ignoring transport energy use, the national welfare surveys typically overestimate energy consumption poverty while underestimating energy expenditure poverty to a substantial degree. Finally, based on a multiple non-linear regression analysis we find, after controlling for spatial sorting of households, that if population densities rise, energy poverty levels in terms of modern fuel use first fall and then increase – thus following a U-shaped pattern. In contrast, the relationship between population density and energy expenditure poverty follows an inverted N-shaped curve. This suggests that urbanization processes in poor countries like Mozambique may not lead to an unequivocal reduction of energy poverty rates, in line with existing evidence on urbanization of poverty in Africa.

1. Introduction

In this article we seek to better understand the potential impact of urbanization in poor countries on the evolution of energy poverty incidence. Energy poverty is generally understood as the lack of access to affordable, reliable, sustainable and modern energy services that are essential to improving quality of life and fostering economic development (see e.g. Ref. [1]Birol 2007 [2],González-Eguino 2015 [3],IEA 2017 [4],Kemmler and Spreng 2007 [5],Saghir 2005 [6],Sovacool 2012). Although energy poverty also occurs in rich countries (see e.g. Ref. [7]Boardman 2009 [8],Brown et al., 2020 [9],Bouzarovski et al., 2012 [10],Thomson et al., 2017), the problem is more widespread, severe and persistent in the global South, and especially in sub-Saharan Africa where it is estimated that currently 53% of the population lack access to electricity and over 80% lack access to clean cooking ([11]IEA 2019). As regards urban growth, in a few decades from now the world urban population is expected to be larger than the entire global population today, and also this trend is largely driven by the unprecedented pace and scale of urbanization in the Global South ([12]Jedwab and Vollrath 2015a [13],United Nations 2018). High urbanization rates, combined with large population bases, make that by 2050 Asia is expected to house more than 50% of the total urban population, while more than half a billion people will be added to Africa's urban population by 2040. What are the implications of this urbanization trend for energy poverty? Does the increasingly urban context help to reduce energy poverty, or can we instead expect urbanization of energy poverty?

Remarkably, the implications of urban growth in the global South on the evolution energy poverty is yet a somewhat under-researched topic. Over the last two decades, in the development studies literature there is a growing realization that although urbanization plays a positive role in overall poverty reduction, the urban share of poverty is rising, which implies 'urbanization of poverty' ([14]Ravallion 2002 [15],Ravaillon et al., 2007). In this literature attention is also paid to the challenge of

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properly measuring urban poverty, in distinction from rural poverty, given the specificities of the urban context (see e.g. Ref. [16]Lucci et al., 2018 [17],Mitlin and Satterthwaite 2013). In the urban economics literature, the relationship between population density and energy use receives growing attention (see, e.g. Ref. [18],Glaeser and Kahn 2010 [19],Larson and Yezer 2015 [20],Sugahara and Bermont 2016). An influential argument here is that urban agglomerations reduce average energy consumption of households because higher population density leads to lower average commuting distance, increasing public transport usage, and smaller housing units. This so-called compact-city argument received, however, only very little attention in the context of developing countries (exceptions include [21]Jenks and Burgess 2004 and [22] Permana et al., 2008) and it does not address the issue of energy poverty as such. Meanwhile, in the rapid growing energy poverty literature much attention is devoted to the integrated measurement of various dimensions of energy poverty (see Ref. [23]Culver 2017 and [24]Pelz et al., 2018 for recent reviews), but these studies do not explicitly consider the role of urbanization or (increasing) population density as a potential driving force of energy poverty incidence. Although studies of energy poverty in poor countries report, as expected, lower levels of energy poverty in cities compared to rural areas (e.g. Ref. [25]Adusah-Poku and Takeuchi 2019), the question as to whether the urban share of energy poverty in those countries is rising – i.e. urbanization of energy poverty – remains underexposed.

This paper aims to contribute to this research gap. We do so by relating for one African country, Mozambique, and two years (2003 and 2015) spatial variation in population density to spatial variation in three commonly used indicators of energy poverty. Insight into the case of Mozambique transcends local and regional importance because we think that the case of Mozambique exemplifies the future development of Africa's energy poverty situation in many respects ([26]Mahumane and Mulder 2016). Like many other African nations, Mozambique is among the poorest countries in the world[1] while at the same time featuring strong economic and population growth and (thus) rapid urbanization. Interestingly, economic growth is partly driven by the development of its large energy reserves, especially natural gas. Over the last decade, the country's average annual population growth has been almost 3%, and the urbanization level increased with about 5% points to 36%. The latter implies that about 3.5 million people have been added to Mozambique's urban population since 2010. Clearly this stimulated an energy demand transition that led people, like in many other poor countries, to switch from consuming traditional biomass to modern energy fuels like electricity, LPG and transport fuels ([27]Mahumane and Mulder 2019) and which impacts energy poverty incidence patterns.

Our analysis is based on two newly constructed datasets with information on energy use and socio-economic characteristics for a representative sample of households, in combination with information on location characteristics like population density, biomass prices, and access to electricity. The first dataset covers households throughout the country, with information originating mainly from two national household welfare surveys (2003 and 2015). The second dataset covers a representative sample of households in the metropolitan area of the capital city Maputo, with data collected by ourselves through an extensive survey (2015). For comparison, we add data on income poverty measured in terms of dollars expenditure per day. The main advantage of the first dataset is of course its national coverage, but an important shortcoming is that its energy data de facto neglect household energy use for (urban) transport, which of course may lead to biased estimates of urban energy poverty. This is typical for national welfare surveys in many developing countries. The second dataset solves for this shortcoming as it contains spatially explicit information on energy use for urban transport.

The article continues as follows. We first explore (in section 2) more in-depth the potential implications of urbanization on the evolution energy poverty in developing countries, based on a review of key insights from several strands of the literature. Next we describe our data (section 3), followed by specifying (in section 4) various energy poverty incidence levels in Mozambique across time and space, distinguishing between rural and urban areas as well the capital city Maputo. Next, we apply (in section 5) a regression analysis to identify the role of population density on energy poverty incidence, after controlling for household and location characteristics. We conclude (in section 6) by discussing the potential implications of our findings and indicating promising avenues for future research into the link between urbanization and energy poverty.

2. Urbanization and energy poverty

From a theoretical perspective, the relationship between urbanization and energy poverty in developing countries is ambiguous in advance. A large body of economic literature on the consequences of urbanization provides strong evidence that an increase in density is associated with welfare enhancing agglomeration effects (see Ref. [30] Combes and Gobillon 2015 [31],Ahlfeldt and Pietrostefani 2017). In other words, clustering of people and firms in urban areas cause cities to act as engines of local, national and global economic growth ([32] Duranton 2008). The implied income growth of urban households is in principle associated with an expected decrease in energy poverty, not only because more purchasing power leads to increasing energy consumption levels but also because the relative availability, affordability, and superiority of modern fuels in urban areas encourages a consumption shift away from traditional fuels (see e.g. Ref. [33]Barnes et al., 2004 [34],Fan et al., 2017 [35],Franco et al., 2017 [36],Parikh and Shukla 1995 [37],Poumanyvong and Kaneko 2010). Indeed, as noted before, energy poverty studies for poor countries frequently report relative low energy poverty levels in urban areas (e.g. Ref. [25]Adusah-Poku and Takeuchi 2019).

But, across developing countries there is also evidence of "urbanization without growth" ([38]Fay and Opal 2000 [29],Jedwab and Vollrath 2015a) and even "urbanization of poverty" ([14]Ravallion 2002 [15],Ravaillon et al., 2007). Several causes can underlie this phenomenon, including rural-urban migration with the rural poor urbanizing faster than the population as a whole, but also rapid natural population growth from within cities. As regards the latter, evidence has been presented that especially across Africa, urban fertility appears to remain high in urban areas while mortality falls ([39]Christiaensen et al., 2013 [40],Jedwab and Vollrath, 2015b). Moreover, an increase in urban density is also associated with congestion forces, including higher housing and food prices and indeed also higher prices of biomass energy sources. As urban areas develop, biomass supplies often are depleted in a larger radius around the city and the implied increased scarcity as well as rising transit cost may then lead to an increase in the relative price of fuelwood and charcoal ([33]Barnes et al., 2004 [41],Sedano et al., 2016). It is well-known that especially charcoal continues to be a very common cooking fuel in cities across low-income countries, for both households with and without electricity access (e.g. Ref. [42]Castán Broto et al., 2020 [43],Zulu and Richardson 2013). There is a predicted 40% rise in biomass energy demand across Africa by 2040 ([3]IEA, 2017), including a rising demand for charcoal in cities, which is directly linked to population growth ([44]Chidumayo & Gumbo, 2013). It therefore cannot be taken for granted that urbanization in the global South will automatically eradicate energy poverty, on the contrary it may even exacerbate the energy poverty problem.

It is exactly this potential ambiguous relationship that is the subject of our study. Hence, our analysis aims to bridge the various strands of literature mentioned above. In contrast to the existing energy poverty literature, we explicitly test for the hypothesis of urbanization of energy

[1] Mozambique ranks 181th out of 189 countries in the latest Human Development Index in 2019 ([28]UNDP 2020). See also [29]Jones and Tvedten (2019) for an excellent portrayal of poverty incidence in Mozambique.

poverty – i.e. the question whether the urban share of energy poverty in developing countries like Mozambique is rising. In contrast to the urban and development economics literatures we analyze the relationship between urbanization and energy poverty in the context of the Global South. Another novelty of our approach is that we combine data from a local urban survey and a typical national welfare survey. Furthermore, our local urban survey purposely includes information on urban transport energy use. This allows us to not only identify the role of transport energy consumption in the observed urban energy poverty patterns but also the potential bias in energy poverty measurement based on national welfare surveys, since the latter usually do not include (reliable) data on transport energy use.

It is widely recognized that energy poverty is a multidimensional phenomenon because access to affordable, reliable, sustainable and modern energy services is a prerequisite for meeting needs as diverse as food, clean water, adequate sanitation, shelter and healthcare as well as education, mobility and mechanical power ([2]González-Eguino 2015 [3],IEA 2017). Hence, no single, universally accepted measure of energy poverty exists that unambiguously defines what it is to be below the energy poverty line. Instead, various definitions and approaches to the measurement of energy poverty have been developed that measure energy in terms of energy access, energy use, energy expenditure, energy quality or energy service outcomes like health effects ([23]Culver 2017). Over the last decade, much attention has been devoted to ways of measuring energy poverty more comprehensively (see, for example [45],Day et al., 2016 [23],Culver 2017 [2],González-Eguino 2015 [24], Pelz et al., 2018 [46],Pachauri and Spreng 2011 [5],Saghir 2005 [47], Sovacool et al., 2012 [48],Wang et al., 2015). The various attempts to conceptualize the multidimensional nature of energy poverty include, amongst others, the development of a so-called access-use framework ([4]Kemmler and Spreng 2007 [49],Pachauri et al., 2004 [50]; Pachauri and Spreng, 2011), the Total Energy Access (TEA) approach ([51,52] Practical Action 2010, 2012), the Multidimensional Energy Poverty Index (MEPI [53,54];Nussbaumer et al., 2012; 2013) and the World Bank's Multitier Framework for Measuring Energy Access (MTF [55]; Bhatia and Angelou 2015). The recent literature on measuring energy poverty includes several applications and proposals for further refinement of these multi-indicator frameworks ([56]Charlier et al., 2021 [57],Middlemiss et al., 2019 [58],Pachauri and Rao 2020 [59],Robinson et al., 2018 [60],Sadath and Acharya 2017 [48],Wang et al., 2015). But, as previously noted, also these studies in general do not yet explicitly consider the specificities of the urban context as a distinguishing determinant of energy poverty incidence, although the framework proposed by Ref. [58]Pachauri and Rao (2020) would in principle allow to do so by considering energy supply conditions as an important poverty incidence criterium.

In contrast, we do not aim to improve measurement of energy poverty in all its complexity, but instead focus on better understanding the link between energy poverty and urbanization. Moreover, as argued, amongst others, by Ref. [23]Culver (2017) [24],Pelz et al. (2018) and [58]Pachauri and Rao (2020), multi-indicator frameworks like MTF are complex to operationalize, set high standards for availability and quality of data and tend to be conceptually unclear because of the need to aggregate and weight various poverty dimensions with different units of measurement. For these reasons, and given the data limitations that we faced in Mozambique, we therefore refrain from engaging with a multi-tier framework. Instead we use a combination of three commonly used binary indicators to measure different dimensions of the energy poverty problem: low energy consumption quantities, high energy expenditure shares, and a lack of access to modern energy types. Their ease of interpretation helps in providing insight as to how energy poverty relates to population density, while the combined use of these indicators helps to assess how population density may relate differently to three different dimensions of the energy poverty problem. In doing so, we find evidence that the relationship between urbanization and energy poverty indeed is ambiguous and depends on which aspect of energy poverty is measured.

3. Data

As already mentioned, our analysis is based on two newly constructed datasets that contain microdata on energy use and a series of socio-economic characteristics for a representative sample of households. Information for the first dataset originates from two national household welfare surveys (2003 and 2015) conducted by Mozambique's National Institute of Statistics (INE) with the aim to compile living conditions ([61,62]INE 2004, 2015). The surveys are conducted during a 12-month period, with a view to reflecting the seasonality during the year. For the sake of representativeness, these surveys are designed using a sampling, probabilistic and stratified approach, with the national population censuses as their basis. We build our energy dataset by first calculating from these national survey data the total expenditures of a household from different consumption categories. Second, we calculated the energy expenditures and consumption quantities for the different energy types used by households. Since unit quantities for biomass fuels (firewood, charcoal and wood residue) were reported in various location specific quantities – including kilograms, bundles, tins, small bags or a standard weighted-sack for the case of charcoal – we had to convert these local units to standard units (kilograms) corrected for fuel efficiency. Third, we constructed unit energy prices by dividing energy expenditures over quantities. Finally, we cleaned the data by controlling for outliers.[2] This procedure yielded a sample of, respectively, 7404 (2003) and 9730 (2015) households of which correspondingly 47.5% (2003) and 50.2% (2015) were situated in urban areas – including 1276 (2003) and 1433 (2015) households living in the Maputo area.

Our second dataset covers a representative sample of 1050 households in the metropolitan area of the capital city Maputo, with information collected by ourselves through an extensive survey (2015). The design of the questionnaire was derived from an already proved city-wide questionnaire survey and accounting methodology for household interviews in Xiamen City in southeast China ([63]Lin et al., 2013). We adapted the questionnaire to the specifics of the Maputo urban area. The survey includes questions about household characteristics and energy consumption for both dwelling and transport services. The full questionnaire is included in the (online) supplemental Annex to this article. As noted before, the second dataset solves for the shortcoming that the energy data from the national welfare surveys de facto neglect household energy use for (urban) transport.

Table 1 summarizes the key characteristics of our data. In addition to the sample characteristics, it shows the percentage usage of the different energy types in our newly constructed dataset, both for the total sample and for the urban-rural stratification. Overall, the percentage of households using electricity, LPG and charcoal increases over time, mainly driven by urban consumption. Especially in the Maputo metropolitan area, LPG is developing into an important source of residential energy use. The use of kerosene is declining in rural and in urban households, most likely because of price increases. Charcoal continues to be an important source of domestic energy for urban households, the percentage of households using it even increases over time. Firewood obviously is the dominant source of energy among rural households, but also continues to be used by urban households. In other words, many poor households rely on multiple energy sources for their energy needs, including electrified households in urban areas. These observations are in line with previous research across developing countries that noticed complex patterns of (urban) energy use including fuel stacking, thus challenging the idea that households transition up an 'energy ladder'

[2] Further details on these procedures, including the energy conversion factors that we applied, are described in the (online) supplemental Annex to this article.

Table 1
Key data characteristics.

		Total		Urban					Rural	
		Mozambique		Mozambique		Maputo			Mozambique	
		INE 2003	INE 2015	INE 2003	INE 2015	INE 2003	INE 2015	Own 2015	INE 2003	INE 2015
Coverage										
Households	#	7404	9730	3518	4882	1276	1433	1050	3886	4848
Urban	%	47.5	50.2	100	100	100	100	100	0	0
Rural	%	52.5	49.8	0	0	0	0	0	100	100
Provinces	#	11	11	–	–	2	2	2	–	–
Districts	#	143	144	27	37	8	9	9	116	107
Percentage of households using a certain energy type										
Electricity	%	12.2	14.7	25.5	27.1	39.1	38.1	95.2	0.2	2.1
LPG	%	2.7	4.4	5.5	8.4	14.3	24.2	51.7	0.1	0.4
Kerosene	%	52.6	5.2	58.4	6.6	51.0	1.2	0	47.4	3.8
Charcoal	%	13.1	30.8	27.0	54.6	35.3	69.2	88.6	0.4	6.8
Firewood	%	67.4	70.5	42.6	43.2	17.6	13.1	13.2	89.8	98.0
Candles	%	7.3	14.7	12.0	24.6	16.4	36.2	0	3.0	4.8
Petrol	%	1.6	0.0	3.0	0.0	5.7	0.0	32.6	0.4	0.0
Diesel	%	2.2	0.0	2.5	0.0	2.1	0.0	8.7	2.0	0.0
Mean values of the key variables underlying our (energy) poverty indicators										
Energy use [a]	GJ	8,4	10,2	10,5	12,8	10,9	13,7	13,6	6,5	7,7
Modern energy use [a]	GJ	1,9	1,7	1,9	1,7	2,6	2,0	8,5	2,2	1,3
Share modern energy use	%	56,2	43,7	56,2	44,9	63,5	51,3	51,1	56,3	26,7
Total expenditure [b]	$	2,4	3,9	4,0	6,3	6,5	9,0	12,2	0,9	1,5
Energy expenditure [b]	$	0,2	0,4	0,3	0,5	0,5	0,9	1,6	0,1	0,2
Share energy expenditure	%	13,2	15,1	10,7	12,1	9,8	13,1	20,6	15,4	18,2

Notes.
[a] Per capita per year.
[b] Per capita per day.

through linear fuel switching (see, for example [64],Aitken 2007 [42], Castán Broto et al., 2020 [65],ESMAP 2000 [66],Munro and Bartlet 2019 [67],Van der Kroon 2013).

To further describe the energy consumption patterns by the households in our datasets, we also summarize in Table 1 the mean values of the key variables underlying our (energy) poverty indicators.[3] In short, the numbers based on INE data show that in 2015 the average quantity of per capita energy use in Mozambique was just over 10 GJ per year of which 1.7 GJ is modern energy fuels. Although this is roughly 20% higher than in 2003, still only about 15% of all households in Mozambique consumed some form of modern energy in 2015. These households are predominantly located in urban areas and they spend about 50% of their energy expenditures on modern fuels. The per capita consumption of modern energy in 2015 is lower than in 2003, especially in rural areas, which of course reflects increased access to modern fuels by poor people (think of rural electrification) who yet consume relatively small quantities of modern fuels. Furthermore, the data reveal that, on average, Mozambican households spent in 2015 close to 4 USD per day, but with substantial variation across space: 1.5 USD in rural areas against 9 USD in Maputo. Average total energy expenditure share was about 15% in 2015, varying between 12.1% in urban areas to 18.2% among rural households.

A comparison between the INE data and our own survey for the capital city Maputo first yields a similar number for total energy use (around 13.6 GJ), which is reassuring. Also, both surveys show that in Maputo modern fuels comprise on average 44–50% of total energy use in those households that use modern energy fuels. In contrast, our own survey for Maputo yields a much higher percentage of households using some form of modern energy (especially electricity) as well as a much higher average level of modern energy use. This is partly due to the fact that our own survey includes more higher income households (see also Fig. 3, bottom-right) and because the INE samples largely ignore the consumption of transport fuels for transport.[4] Finally, our own survey data contain a higher expenditure share for energy use than INE data (20% versus 13%). If we exclude transport fuel expenditures, however, our survey data yield a comparable total energy expenditure share value (almost 15%), which suggests that energy expenditures for transport in Maputo are in the range of 5–6%. In the next section we assess how these observations translate into energy poverty incidence levels across urban and rural areas and across the two samples for Maputo.

4. Energy poverty incidence

To measure energy poverty incidence levels we define energy poverty lines for each of our three indicators of energy poverty incidence: low energy consumption quantities, high energy expenditure shares, and lack of access to modern energy types. Energy poverty incidence is then defined as the fraction of the total population in a district that falls below the relevant specified threshold. In doing so, we distinguish between weak and strong poverty. In Table 2 we summarize the thresholds that we have chosen for weak and strong poverty.

We arrived at these values on the basis of an assessment of several

Table 2
Poverty thresholds.

Poverty indicator		Threshold	
		Weak Poverty	Strong Poverty
Energy consumption (cap/year)	EP_{GJ}	<6 GJ	<4 GJ
Energy expenditure share	EP_{exp}	>10%	>20%
Modern energy share	EP_{modern}	<10%	<4%
Income per capita per day	IP	<4 USD	<2 USD

[3] We refer to part B of the Annex to this article for a more detailed presentation of the accompanying summary statistics.

[4] Underlying data reveal that transport fuels make up for about 50% of all modern energy use by households in Maputo.

access-use poverty lines developed in the literature over time, in combination with the actual data values in our sample and the context of Mozambique. For example, the IEA in its World Energy Outlook 2009 [11] identified three levels of access to energy services depending on household energy needs and the benefits that energy services provide [68].Vermaak et al. (2009) use these levels to develop energy poverty lines for South Africa. They include the minimum level of energy access required to i) satisfy basic human needs (2.4 GJ), ii) improve productivity (4.3 GJ), and iii) satisfy modern society needs (7.2 GJ). In his recent review of energy poverty measures [2] González-Eguino (2015) indicates as values for these three thresholds, 5/5–10/10-25 GJ/cap year, respectively. For defining energy poverty in terms of expenditure, we go back to the well-known Ten-Percent-Rule (TPR): households are energy poor if they (have to) spend more than ten percent of their net income on adequate energy services. We use the same rule as benchmark for defining energy poverty in terms of modern fuel share. Finally, to measure income poverty we draw upon the widely-use poverty headcount ratios of USD 1.90 and 3.20 (PPP) a day. Admittedly, the threshold values in Table 2 inevitably have a certain arbitrary character and can therefore be disputed. Nevertheless, we think these threshold values are adequate given the purpose of our study and because they balance information from the literature and the actual features and context of our sample, while the differentiation between the weak and strong energy poverty thresholds across three dimensions of energy poverty provides sufficient guarantee that the conclusions of our analysis are not obfuscated by one incorrectly chosen threshold value.

Using these (energy) poverty definitions, we first summarize in Table 3 key facts on the evolution of energy poverty incidence levels between 2003 and 2015 for the country as a whole. The Table first shows that energy poverty levels are high in Mozambique: based on a weak poverty definition, in 2015 around 50% of the population is energy poor in terms of energy consumption and energy expenditure, while around 90% is energy poor in terms of access to modern energy types. If we apply a strong definition of poverty, still around 30% of the Mozambican population is poor in terms of energy use and expenditure, while the overall poverty rate in terms of access to modern energy types remains virtually the same except that it decreases in urban areas. In comparison, income poverty incidence in 2015 was almost 70% for the weak poverty line and 55% for the strong poverty line. As expected, all these national poverty incidence levels are lower in urban areas and higher in rural areas. What is striking, however, is that in rural areas income poverty is more severe than energy poverty, while broadly the opposite is true in urban areas (except for strong energy expenditure poverty).

As regards the dynamics of energy poverty incidence levels, Table 3 shows that energy poverty is gradually decreasing over time, except for energy poverty measured in terms of energy expenditure shares (EP_{exp}). The latter type of energy poverty increases over time, and especially in terms of strong poverty. Clearly, there is a considerable group of people that need more than 20% of their income to meet their energy demand and their share is growing substantially (up to about 37% of the rural and 19% of the urban population in 2015). Although, the urban areas thus still do relatively well as compared to rural areas, Table 3 reveals that also in urban areas this type of energy poverty undeniably is a significant and rapid growing problem.

In contrast, energy poverty measured in terms of consumption quantities (EP_{GJ}) decreases considerably over time, also in urban areas. This implies that the observed rise in energy expenditure poverty is likely to be principally driven by rising energy prices. Finally, energy poverty measured in terms of access to modern energy fuels (EP_{modern}) decreases only slightly over time. The reduction in EP_{modern} is bigger in urban than in rural areas, despite the substantial progress that has been made in electrifying rural areas since 2003. In comparison, income poverty (IP) in terms of per capita expenditure per day decreased substantially since 2003 and especially in urban areas.

To better understand the distributional dynamics that underlie the observed evolution of energy poverty incidence in Mozambique we constructed for each poverty indicator the incidence curve, also known as the Cumulative Distribution Function (CDF). They are presented in Fig. 1 and plot the degree of poverty inequality across the population: each point on the curve shows the total proportion of the population (the vertical axis) that 'achieves' a certain value of, respectively, energy use, energy expenditure share, modern fuel share and income (the horizontal axis). In the figure we also indicate the respective weak and strong poverty lines and we distinguish between the total population (left) and the urban population (right).

Fig. 1 first confirms that energy poverty measured in terms of energy consumption decreases over time and is relatively low for urban households. In addition, it can be seen that the reduction of energy poverty measured in terms of energy consumption has been more or less consistent across the entire population distribution while both urban and total energy poverty incidence levels are similar at very low levels of energy consumption (<4 GJ), i.e. for strong energy poverty levels. In fact, strong energy consumption poverty is slightly more prevalent in urban areas, which suggests that cities are home to a group of people that suffer from deep poverty. In contrast, at higher consumption levels (>4 GJ), poverty incidence in terms of energy use is relatively low in urban areas as compared to the country in total. The dynamics in the energy consumption distribution are highest at low levels of consumption (<4 GJ), whereas at the upper end of the distribution there is relatively little dynamics over time. As regards poverty in terms of energy expenditure share, the Figure reveals that the overall increase of this type of poverty is mainly driven by those households that spent 10–30% of their income on energy while in urban areas, the increase in this type of energy poverty is concentrated among households that spend 10–20% of their income on energy.

As regards access to modern energy use, Fig. 1 shows not only that the share of population with no or very low modern energy consumption has decreased (i.e. energy poverty reduction), but also that the percentage of the population with a high share of modern energy is increasing over time. This pattern is more pronounced in urban areas,

Table 3
Poverty incidence levels over time.

Poverty lines		% of Population						% Change		
Definition		Total		Urban		Rural		Total	Urban	Rural
Weak Poverty		2003	2015	2003	2015	2003	2015	2003–15		
EP_{GJ}	<6 GJ Energy consumption/pp/year	59.1	48.5	54.9	43.4	62.9	53.5	−18.0	−20.9	−15.0
EP_{exp}	>10% Energy expenditure share	48.9	54.9	40.2	45.1	56.8	64.7	12.2	12.2	13.8
EP_{modern}	<10% Modern energy share	91.1	89.6	81.5	80.3	99.7	98.8	−1.6	−1.5	−0.9
IP	<4 $PPP income per capita per day	82.8	68.3	66.4	42.7	97.7	93.9	−17.5	−35.7	−3.8
Strong Poverty										
EP_{GJ}	<4 GJ Energy consumption/pp/year	46.0	32.6	45.1	32.8	46.8	32.5	−29.0	−27.2	−30.6
EP_{exp}	>20% Energy expenditure share	21.9	28.1	15.1	18.7	28.1	37.5	28.5	24.1	33.7
EP_{modern}	<4% Modern energy share	89.5	86.0	78.2	73.9	99.7	98.0	−3.9	−5.5	−1.7
IP	<2 $PPP income per capita per day	69.3	54.7	43.7	26.2	92.6	83.1	−21.1	−39.9	−10.2

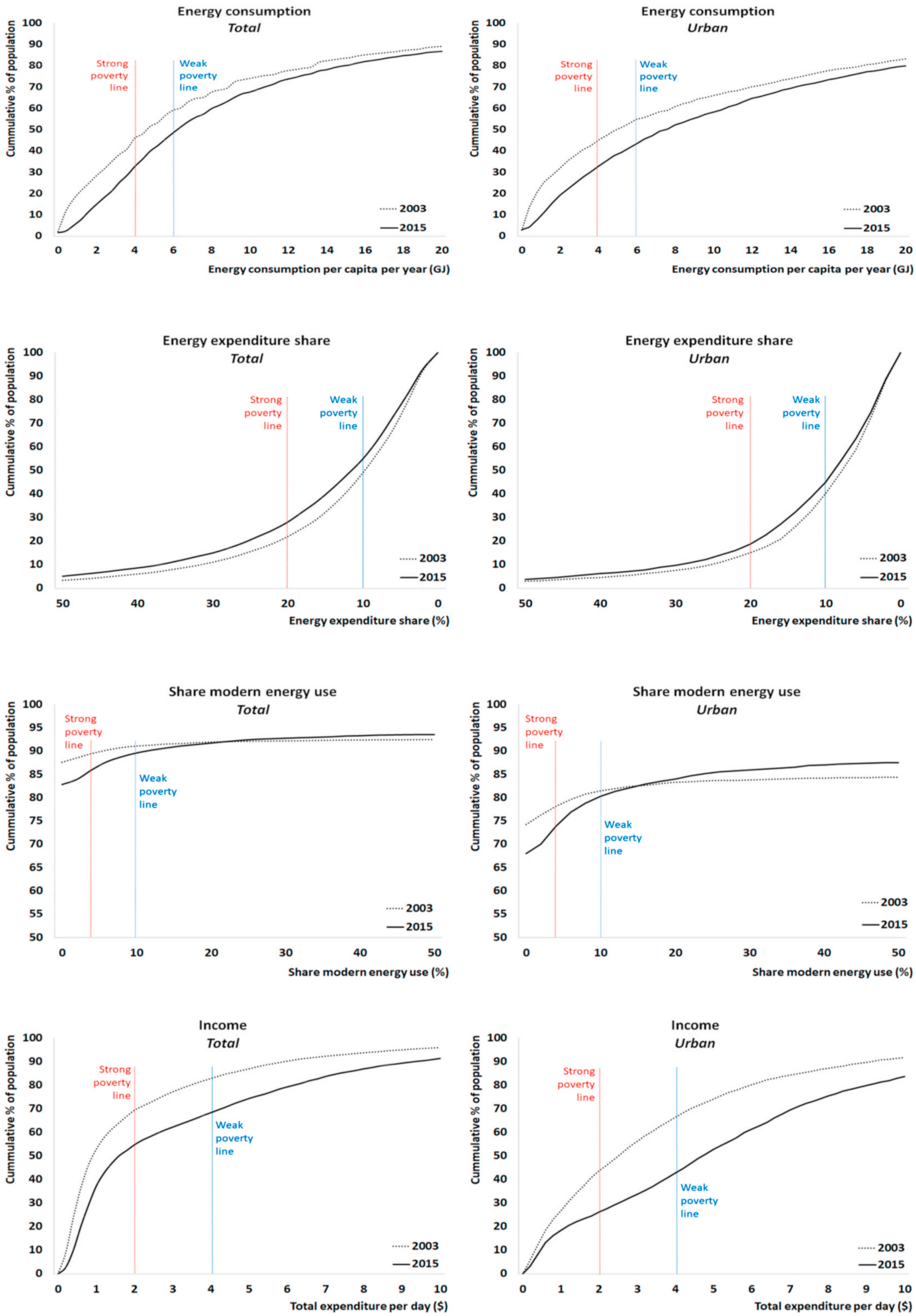

Fig. 1. Incidence curves for energy and income poverty - Mozambique.

where this type of energy poverty of course is relatively low.

Nevertheless, as previously observed, also in urban areas the share of modern fuels in total energy consumption remains low. Finally, Fig. 1 shows that the strong reduction in income poverty in urban areas involves almost all households across the income distribution except the very poor that spend less than 1$ a day. The Figure confirms that the substantial reduction in income poverty correlates with a reduction in energy poverty in terms of (modern) energy use but not with energy poverty in terms energy expenditure. Taken together, the evidence presented thus far suggests that the rising incomes in cities lead to increasing energy use and facilitate a gradual transition towards consuming modern fuels, and that the implied increase in energy use inevitably entails an increase in energy expenditure poverty.

To gain further insight in the urban dimension of the observed energy poverty patterns we use our second dataset for Maputo, based on information from our own survey data. We do so first by presenting in

Table 4
Poverty incidence levels over time – Maputo in comparison to all urban areas.

Poverty lines		% of Population						% Change	
		National Surveys (INE)				Own Survey		National Surveys (INE)	
		2003		2015		2015		2003–15	
Definition		All urban[a]	Maputo	All urban[a]	Maputo	Maputo		All urban[a]	Maputo
Weak Poverty		Total		Total		Dwelling	Total	Total	
EP_GJ	<6 GJ Energy consumption pp/year	54.9	55.3	43,4	39.6	31.1	22.4	−20.9	−28.3
EP_exp	>10% Energy expenditure share	40.2	41.2	45.1	54.2	59.0	73.8	12.2	31.5
EP_modern	<10% Modern energy share	81.5	70.0	80.3	62.9	17.7	16.4	−1.5	−10.1
IP	<4 USD income per capita per day	66.4	40.6	42.7	31.5	–	42.2	−35.7	−44.1
Strong Poverty									
EP_GJ	<4 GJ Energy consumption/pp/year	45.1	46.6	32,8	31.5	11.0	8.51	−27.2	−32.5
EP_exp	>20% Energy expenditure share	15.1	12.4	18.7	21.3	27.0	42.4	24.1	71.9
EP_modern	<4% Modern energy share	78.2	65.8	73.9	54.4	7.2	6.8	−5.5	−17.3
IP	<2 USD income per capita per day	43.7	19.3	26.2	15.1	–	15.2	−39.9	−43.7

[a] Same data as in Table 3.

Table 4 key facts on energy poverty incidence levels in Maputo in 2015, in comparison with the results based on the INE data for Maputo as well as all the previously used INE data for all urban areas (for the sake of comparability, we repeat the latter results from Table 3). Table 4 leads to the following conclusions. First, as expected, in 2015 energy poverty incidence in terms of energy use and especially modern energy use are, like income poverty, substantially lower in Maputo than in the other urban areas in Mozambique on average. Second, the opposite is true for energy poverty in terms of energy expenditure shares, which suggests that the largest city in Mozambique indeed is also the most expensive city in terms of energy consumption. Third, like for the country as a whole, income and energy poverty incidence levels in Maputo decrease over time, except for energy poverty in terms of energy expenditure shares. But these changes are more pronounced for Maputo: the decrease in strong income poverty incidence and energy poverty incidence in terms of energy use (EP_{GJ}) and use of modern fuels (EP_{modern}) is faster in Maputo than in the other urban areas on average; in contrast urban energy poverty in terms of energy expenditure shares (EP_{exp}) increases faster in Maputo than in other urban areas. Fourth, a within-Maputo comparison of energy poverty incidence levels based on INE data and our own survey data reveals that INE data lead to a relatively high EP_{GJ} and EP_{modern}, while the opposite holds for EP_{exp}. This difference is partly due to the fact that the INE data ignore the substantial role of transport fuel consumption on poverty incidence and partly to sampling differences, as previously noted. Consequently, in terms of EP_{modern} our own survey data indeed yield a much lower energy poverty level than INE data: approximately 17% (7%) versus 63% (54%) for weak (strong) poverty. Nevertheless, our survey data confirm the result from the INE data that the group of people living in extreme poverty amounts to about 15% of the population in Maputo.

The role of transport fuel consumption in estimating urban energy poverty can best be illustrated by comparing the fourth and fifth column

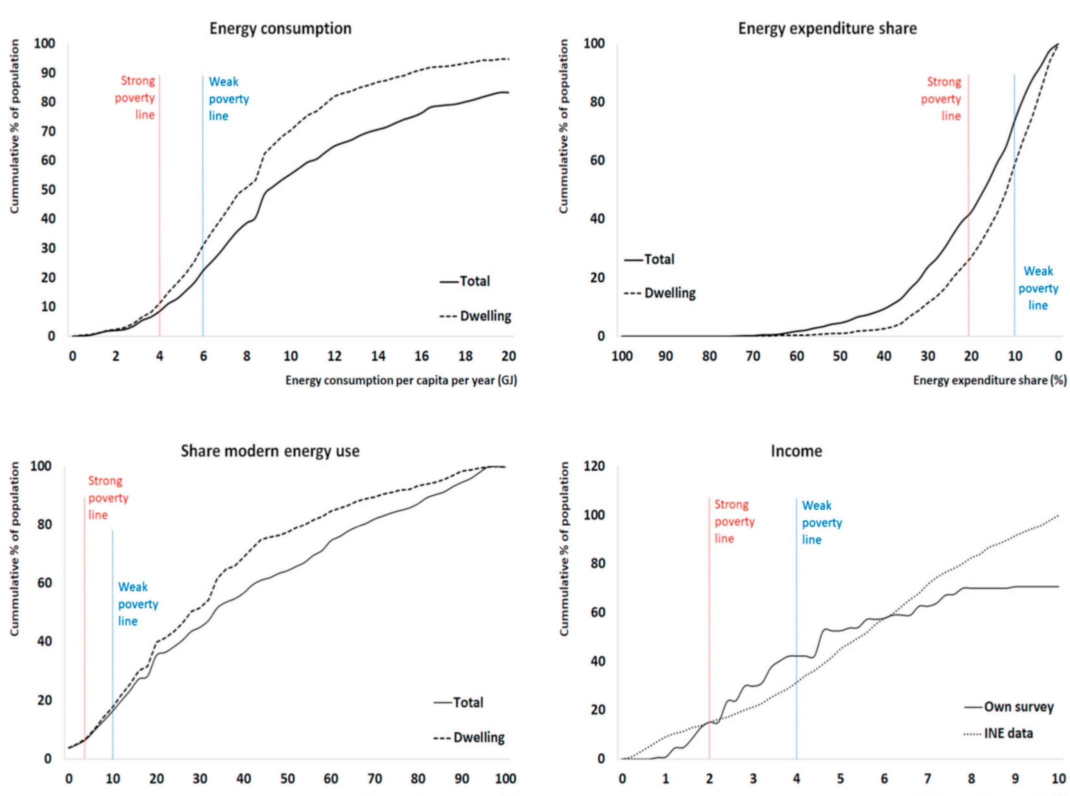

Fig. 2. Incidence curves for energy and income poverty – Maputo.

in Table 4: they show that the weak energy poverty levels for EP_{GJ} and EP_{exp} based on INE data and our own survey are very similar, insofar our own survey data only take dwelling energy consumption into account (approximately 31–39% for EP_{GJ} and 54–59% for EP_{exp}). This is reassuring, because these two series both essentially measure dwelling energy consumption. If we do consider transport fuel use, however, EP_{GJ} decreases from about 31% to 22% for EP_{GJ} and increases from about 59% to almost 74% for EP_{exp} (sixth column in Table 4). In other words, ignoring energy consumption for transport leads roughly to a 10% point overestimation of energy consumption poverty and a 18% point underestimation of energy expenditure poverty.

We conclude this section by showing in Fig. 2 for Maputo in 2015 the various (energy) Cumulative Distribution Functions (CDF) or poverty incidence curves, based on our own survey data. We compare two curves: dwelling energy use and total energy use (the latter thus including transportation energy). For comparison, we also plot the income poverty curve for Maputo based on INE data. A comparison of these two income poverty incidence curves in the bottom-right picture in Fig. 2 reveals that our own survey sample includes relatively less extremely poor as well as very rich households, but relatively more 'middle class' households (2–6 US$ per day). The top-left picture in Fig. 2 shows that the gap between the incidence curves for total and dwelling energy use is largest for households that consume more than 9 GJ per year. In other words, for these households, transport energy use contributes relatively much in lowering this type of energy poverty. The top-right picture in Fig. 2 shows that in terms of energy expenditure poverty, transport fuel expenditures has most impact on poverty incidence levels for those households that spend 15–30% of their income on energy services. Finally, the bottom-left picture shows that especially for the group of households with a modern fuel share of 40–70%, transport fuels make a difference in adequately measuring energy poverty.

Fig. 3. Predicted values for weak (left) and strong (right) poverty incidence levels as function of population density, across districts for 2003 and 2015.

5. Energy poverty and population density

To further explore the relationship between urbanization and energy poverty we develop in this section a multiple non-linear regression framework to statistically test for the hypothesis that population density impacts energy poverty incidence. The basic idea behind our multiple regression strategy is that we need to disentangle the potential role of population density from other variables that also may impact energy poverty incidence and are related to population density. For example, it is quite conceivable that households are not randomly distributed over space – higher educated people with higher incomes and relatively small families are more likely to be found in high density areas (i.e. cities) and therefore less energy poverty is expected in these areas. The same applies to the availability of modern energy and traditional energy sources over space: in a poor country like Mozambique, LPG supply is largely restricted to the urban areas while, conversely, charcoal and fuelwood are more readily available and thus cheaper in rural areas with low population density. In our regression analysis, we therefore first control for the potential impact that such location characteristics as well as the spatial sorting of households may have on energy poverty incidence. Subsequently, we then add population density as an additional variable in our regression framework to analyze whether and how it may have a separate effect on different types of energy poverty, beyond its implicit role through other location characteristics and spatial sorting of households.

To this aim, we exploit cross-district variation in population density to explain both the 2003 and 2015 cross-district variation in energy and income poverty incidence (as measured by our various indicators), while controlling for spatial variation in household characteristics and several other spatially explicit variables. More specifically, we regress per district *i* the (energy) poverty incidence level (*Pov*) on a constant α, a set of *J* control variables (*C*) indexed *j*, a set of *K* spatial variables (*S*) indexed *k*, and population density (*D*) to estimate β, γ and δ, according to:

$$Pov_i = \alpha + \sum_{j=1}^{J} \beta_j C_i^j + \sum_{k=1}^{K} \gamma_k S_i^k + \delta_i D_i + \varepsilon_i, \tag{1}$$

where ε_i denotes a robust standard error clustered per province, to control for unobserved between-province heterogeneity biases by accounting for within-province correlation or heteroscedasticity.

Our key spatial variable of interest in the regression analyses is the population density *D* of a household's district; our data encompass all 143 districts of the country. Our set of control variables *C* include per capita income (expenditure), education level, household size, share of modern energy (except for the case of energy poverty in terms modern energy share), energy prices and whether a households' district has access to electricity. The set of spatial variables *S* includes a dummy for whether a household lives in an urban area in general or the Maputo area in particular. Energy prices are defined in terms of a price index for traditional fuels, i.e. charcoal and fuelwood. We do not control for modern fuel prices because of their limited availability across districts (especially LPG in both years, and electricity in 2003) and lack of spatial variation since electricity prices are determined at the national level and thus identical across the country. Following the definition used by INE, education level is defined as the average education level of all household members above five years old.

Because there is no a priori reason to believe that the relationship between energy poverty and population density should be linear, we explicitly do allow for a non-linear effect in our regression approach. Specifically, we tested for a second degree polynomial function (a U-shape) as well as a third degree polynomial function (a N-shape) in terms of population density (*D*), according to the following two non-linear variants of equation (1):

$$Pov_i = \alpha + \sum_{j=1}^{J} \beta_j C_i^j + \sum_{k=1}^{K} \gamma_k S_i^k + \delta_i D_i + \delta_i {D_i}^2 + \varepsilon_i \tag{1b}$$

$$Pov_i = \alpha + \sum_{j=1}^{J} \beta_j C_i^j + \sum_{k=1}^{K} \gamma_k S_i^k + \delta_i D_i + \delta_i D_i^2 + \delta_i D_i^3 + \varepsilon_i \tag{1}$$

in which the other variables have the same interpretation as in equation (1). We ran separate regressions for each of our (energy) poverty indicators, for both the 2003 and 2015 datasets and distinguishing between weak and strong poverty. In short, the key results from the regression analysis for the strong poverty case resemble, by and large, those for the weak poverty case. Because of space limitations, we therefore only present the regression results for the case of weak poverty and we do so in Tables 5–8, distinguishing different model versions that vary in terms of included variables; we refer to Part C of the Annex to this article for the regression results in case of strong poverty.

We start by presenting in Table 5 the regression results for weak income poverty (IP). First, the results show, as expected, that income poverty incidence correlates with a low expenditure per capita, a low education level, and a large household size. As regards the spatial variables, we find that, as expected, income poverty is lower in urban areas. Interestingly, in 2003 in Maputo income poverty is statistically significantly lower than in other districts, whereas this effect deceases to exist in 2015. Most importantly, if we control the regressions for population density, we find evidence for a non-linear, inverted N-shaped, relationship between income poverty and population density (regression model 5): if a districts population density increases, income poverty first falls, then increases, to fall again at very high levels of population density (i.e. in the largest cities).

In Table 6 we present the results of the same regression framework to explain energy poverty in terms of energy consumption levels (EP_{GJ}). As expected, we find that energy poverty in terms of energy consumption levels is negatively correlated with per capita expenditure but we do not find a statistically significant effect of household size. Interestingly, we find a statistically significant positive effect of education level and a households' share of modern energy use in 2015. We cannot explain the former effect, but the latter effect suggests that at least part of the households may compensate modern energy use with lower levels of use. We also added a quadratic term for modern energy share in the regressions to test for the hypothesis that energy consumption poverty first increases at low levels of modern energy use to decrease at higher levels of modern energy use. In other words, we test whether the potential compensation of modern energy use with lower consumption levels maybe a phenomenon that prevails among (poor) households (presumably at an initial adoption stage) that can only (yet) afford low levels of modern energy use. We did not, however, find any statistical evidence for such a non-linear effect and therefore dropped this version of the regression model. In addition, the regression results show that in 2003 a relatively high energy price for traditional biomass contributes to energy consumption poverty, but this effect was not found any more in 2015. Furthermore, energy consumption poverty incidence levels are lower in electrified districts, but this effects is only statistically significant for the weak poverty case in 2015. As regards the spatial effects, we did not find evidence that energy consumption poverty is statically significantly different in urban areas in general, but we found weak evidence that this type of energy poverty is higher in the Maputo urban area in 2015. Finally, and most importantly, Table 6 shows that we do not find any robust evidence for an independent relationship between population density and energy poverty in terms of energy consumption levels. There is some evidence, however, that energy poverty in terms of energy consumption is relatively high in the Maputo area in 2015.

In Table 7 we present the regression results for the case of energy poverty in terms of energy expenditure share (EP_{exp}). Again, we allowed for a non-linear effect of a households' share of modern energy use on energy poverty to examine the hypothesis that part of the households struggle to pay for their modern energy use consumption, depending on their modern energy use level. The results show that this indeed is the case: if a households' modern energy use increases, EP_{exp} first falls in

Table 5
Explaining weak income poverty (IP).

	Model 1		Model 2		Model 3		Model 4		Model 5	
	2003	2015	2003	2015	2003	2015	2003	2015	2003	2015
Log (expenditure per capita)	−0.122***	−0.111***	−0.100***	−0.100***	−0.067***	−0.090***	−0.059***	−0.080***	−0.053***	−0.071***
	(0.023)	(0.018)	(0.020)	(0.016)	(0.015)	(0.017)	(0.013)	(0.016)	(0.012)	(0.014)
Education	−0.109***	−0.187***	−0.090***	−0.160***	−0.066***	−0.165***	−0.043***	−0.142***	−0.044***	−0.147***
	(0.021)	(0.021)	(0.021)	(0.021)	(0.013)	(0.019)	(0.013)	(0.024)	(0.013)	(0.019)
Household size	−0.037***	−0.037***	−0.032***	−0.031***	−0.018**	−0.026**	−0.014**	−0.022**	−0.011**	−0.018**
	(0.012)	(0.011)	(0.010)	(0.010)	(0.007)	(0.010)	(0.006)	(0.010)	(0.005)	(0.009)
Spatial effects:										
Dummy Urban area			−0.087***	−0.127***						
			(0.028)	(0.030)						
Dummy Maputo					−0.413***	0.266	−0.197	0.338	−0.181	−0.028
					(0.078)	(0.308)	(0.172)	(0.222)	(0.128)	(0.044)
Population Density					−0.031	−0.623*	−0.822**	−1.708***	−1.475***	−2.946***
					(0.042)	(0.326)	(0.359)	(0.299)	(0.379)	(0.517)
Population Density^2							0.359**	1.052***	1.598***	7.225***
							(0.148)	(0.293)	(0.461)	(2.076)
Population Density^3									−0.478***	−4.552***
									(0.159)	(1.575)
Constant	0.946***	1.261***	0.968***	1.219***	0.949***	1.201***	0.932***	1.161***	0.941***	1.176***
	(0.081)	(0.066)	(0.067)	(0.061)	(0.052)	(0.059)	(0.049)	(0.062)	(0.044)	(0.056)
# Observations	134	141	134	141	134	141	134	141	134	141
R^2	0.603	0.726	0.642	0.769	0.816	0.762	0.842	0.788	0.862	0.813

Standard errors are in parenthesis; ***p < 0.01, **p < 0.05, *p < 0.1.

Table 6
Explaining weak energy poverty in terms of energy use (EP_GJ).

	Model 1		Model 2		Model 3		Model 4		Model 5	
	2003	2015	2003	2015	2003	2015	2003	2015	2003	2015
Log (expenditure per capita)	−0.227***	−0.158***	−0.228***	−0.158***	−0.221***	−0.156***	−0.221***	−0.153***	−0.221***	−0.153***
	(0.034)	(0.017)	(0.036)	(0.017)	(0.034)	(0.019)	(0.034)	(0.019)	(0.034)	(0.019)
Education	0.089**	0.002	0.088**	0.002	0.090**	0.006	0.089**	0.011	0.089**	0.012
	(0.039)	(0.017)	(0.043)	(0.017)	(0.040)	(0.017)	(0.043)	(0.019)	(0.044)	(0.019)
Household size	0.026	0.010	0.025	0.010	0.028	0.012	0.028	0.014	0.028	0.014
	(0.017)	(0.017)	(0.018)	(0.017)	(0.018)	(0.018)	(0.018)	(0.018)	(0.019)	(0.019)
Share modern energy use	0.005	0.004***	0.005	0.004***	0.007*	0.004***	0.007	0.004***	0.007*	0.004***
	(0.005)	(0.001)	(0.005)	(0.001)	(0.004)	(0.001)	(0.005)	(0.001)	(0.004)	(0.001)
Energy price	0.150**	0.001	0.150**	0.001	0.152**	0.001	0.152**	0.001	0.152**	0.001
	(0.061)	(0.001)	(0.060)	(0.001)	(0.063)	(0.001)	(0.064)	(0.001)	(0.064)	(0.001)
Dummy Electrification	−0.012	−0.206**	−0.012	−0.206**	−0.011	−0.200**	−0.012	−0.194**	−0.012	−0.194**
	(0.031)	(0.081)	(0.031)	(0.080)	(0.031)	(0.082)	(0.032)	(0.082)	(0.033)	(0.083)
Spatial effects:										
Dummy Urban area			0.004	0.003						
			(0.046)	(0.029)						
Dummy Maputo					−0.022	0.131	−0.034	0.145*	−0.034	0.129*
					(0.062)	(0.084)	(0.077)	(0.083)	(0.080)	(0.076)
Population Density					−0.046*	−0.166	0.009	−0.494	0.019	−0.561
					(0.025)	(0.113)	(0.420)	(0.339)	(0.599)	(0.447)
Population Density^2							−0.025	0.321	−0.048	0.665
							(0.184)	(0.235)	(0.614)	(1.002)
Population Density^3									0.009	−0.258
									(0.182)	(0.646)
Constant	−0.354**	0.427***	−0.355**	0.427***	−0.356**	0.411**	−0.354**	0.393**	−0.354**	0.391**
	(0.162)	(0.152)	(0.162)	(0.152)	(0.162)	(0.158)	(0.167)	(0.161)	(0.168)	(0.163)
# Observations	133	141	133	141	133	141	133	141	133	141
R^2	0.405	0.490	0.405	0.490	0.408	0.493	0.408	0.497	0.408	0.497

Standard errors are in parenthesis; ***p < 0.01, **p < 0.05, *p < 0.1.

order to increase at higher levels of modern energy use. In other words, urban people appear to face a trade-off between the different dimensions of energy poverty. Furthermore, we find that EP_{exp} correlates negatively with lower levels of education and larger household size but positive with higher expenditure levels (income). The former two effects are in line with expectations. The latter effect is surprising at first sight, but falls in line with the result that EP_{exp} increases at higher levels of modern energy use; it can probably best be understood as that higher incomes are relatively dependent on modern energy types to meet their energy demand, which implies relatively high energy costs. Furthermore, Table 7 shows that we could not find a statically significant effect of the energy price for traditional biomass and whether or not a district is electrified. As regards the spatial effects, we find evidence that EP_{exp} is lower in urban areas in general, whereas it is higher in the Maputo area. In addition, we find evidence for a non-linear, relationship between EP_{exp} and population density: if a districts population density increases, energy poverty in terms of expenditure share first falls in order to increases at increasing levels of population density (regression model 4) - implying a U-shaped pattern. In addition, for 2015 we find evidence that energy expenditure poverty may actually fall again at very high levels of population density (i.e. in the largest cities), thus follows an inverted N-shaped pattern like income poverty (regression model 5). This latter

Table 7
Explaining weak energy poverty in terms of energy expenditure share (EP_exp) – including non-linear effect of share modern energy use.

	Model 1		Model 2		Model 3		Model 4		Model 5	
	2003	2015	2003	2015	2003	2015	2003	2015	2003	2015
Log (expenditure per capita)	0.209***	0.200***	0.217***	0.206***	0.210***	0.202***	0.221***	0.210***	0.221***	0.212***
	(0.036)	(0.025)	(0.037)	(0.024)	(0.037)	(0.025)	(0.037)	(0.025)	(0.037)	(0.025)
Education	−0.196***	−0.081**	−0.186***	−0.063	−0.195***	−0.062	−0.178***	−0.064	−0.178***	−0.055
	(0.046)	(0.040)	(0.049)	(0.039)	(0.047)	(0.042)	(0.049)	(0.042)	(0.049)	(0.043)
Household size	−0.049**	−0.027	−0.048*	−0.024	−0.049*	−0.024	−0.044*	−0.019	−0.044*	−0.018
	(0.025)	(0.025)	(0.025)	(0.025)	(0.025)	(0.026)	(0.025)	(0.026)	(0.025)	(0.026)
Share modern energy use	−0.131**	−0.072**	−0.119**	−0.069**	−0.137**	−0.089**	−0.122**	−0.067*	−0.104*	−0.073**
	(0.051)	(0.030)	(0.050)	(0.028)	(0.059)	(0.039)	(0.047)	(0.035)	(0.060)	(0.036)
Share modern energy use^2	0.005***	0.001**	0.004**	0.001**	0.005**	0.001**	0.005***	0.001**	0.004*	0.001**
	(0.002)	(0.000)	(0.002)	(0.000)	(0.002)	(0.000)	(0.002)	(0.000)	(0.002)	(0.000)
Energy price	0.019	−0.001	0.015	−0.001	0.021	−0.001	0.023	−0.001	0.022	−0.001
	(0.029)	(0.001)	(0.029)	(0.001)	(0.029)	(0.001)	(0.028)	(0.001)	(0.028)	(0.001)
Dummy Electrification	−0.011	−0.124	−0.004	−0.133	−0.010	−0.109	−0.001	−0.097	−0.001	−0.090
	(0.037)	(0.131)	(0.038)	(0.129)	(0.037)	(0.133)	(0.038)	(0.133)	(0.039)	(0.134)
Spatial effects:										
Dummy Urban area			−0.047	−0.088**						
			(0.053)	(0.040)						
Dummy Maputo					−0.148	0.356*	0.107	0.366***	0.103	0.214**
					(0.093)	(0.182)	(0.159)	(0.129)	(0.152)	(0.095)
Population Density					0.051	−0.363*	−0.841**	−1.348***	−1.115*	−2.003***
					(0.080)	(0.218)	(0.389)	(0.367)	(0.591)	(0.516)
Population Density^2							0.441**	0.955***	1.003	4.333***
							(0.190)	(0.271)	(0.791)	(1.359)
Population Density^3									−0.226	−2.537***
									(0.292)	(0.916)
Constant	1.700***	1.427***	1.698***	1.415***	1.703***	1.383***	1.672***	1.351***	1.658***	1.334***
	(0.159)	(0.242)	(0.160)	(0.235)	(0.161)	(0.245)	(0.159)	(0.246)	(0.162)	(0.248)
# Observations	133	141	133	141	133	141	133	141	133	141
R^2	0.325	0.441	0.328	0.461	0.329	0.453	0.342	0.472	0.342	0.477

Standard errors are in parenthesis; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table 8
Explaining weak energy poverty in terms of share modern energy use (EP_modern).

	Model 1		Model 2		Model 3		Model 4		Model 5	
	2003	2015	2003	2015	2003	2015	2003	2015	2003	2015
Log (expenditure per capita)	−0.047***	−0.019**	−0.036**	−0.017***	−0.030***	−0.005	−0.016*	−0.000	−0.014	−0.003
	(0.014)	(0.007)	(0.015)	(0.006)	(0.010)	(0.006)	(0.009)	(0.006)	(0.009)	(0.005)
Education	−0.094***	−0.118***	−0.082**	−0.114***	−0.080***	−0.098***	−0.042***	−0.088***	−0.041***	−0.086***
	(0.031)	(0.020)	(0.034)	(0.021)	(0.024)	(0.011)	(0.013)	(0.010)	(0.013)	(0.008)
Household size	−0.020***	−0.007	−0.018***	−0.006	−0.014***	0.001	−0.008***	0.003	−0.007***	0.001
	(0.005)	(0.007)	(0.005)	(0.006)	(0.004)	(0.005)	(0.003)	(0.005)	(0.003)	(0.005)
Energy price	−0.010	−0.000	−0.013	−0.000*	−0.005	−0.000	−0.005	−0.000	−0.004	−0.000
	(0.014)	(0.000)	(0.013)	(0.000)	(0.013)	(0.000)	(0.009)	(0.000)	(0.009)	(0.000)
Dummy Electrification	−0.036***	−0.038	−0.028**	−0.040	−0.035***	−0.022	−0.025**	−0.014	−0.022**	−0.019
	(0.012)	(0.061)	(0.012)	(0.064)	(0.011)	(0.053)	(0.010)	(0.048)	(0.011)	(0.047)
Spatial effects:										
Dummy Urban area			−0.046**	−0.018						
			(0.023)	(0.013)						
Dummy Maputo					−0.153*	−0.193**	0.175*	−0.161*	0.188**	−0.039
					(0.086)	(0.088)	(0.100)	(0.095)	(0.094)	(0.045)
Population Density					−0.003	−0.040	−0.524***	−0.850***	−1.473***	−0.105
					(0.049)	(0.110)	(0.173)	(0.203)	(0.400)	(0.265)
Population Density^2							0.403***	0.467***	0.989*	−1.610
							(0.106)	(0.154)	(0.548)	(1.273)
Population Density^3									−0.166	1.530
									(0.201)	(1.006)
Constant	1.093***	1.267***	1.100***	1.264***	1.086***	1.196***	1.054***	1.167***	1.053***	1.168***
	(0.053)	(0.092)	(0.049)	(0.096)	(0.046)	(0.075)	(0.038)	(0.068)	(0.038)	(0.067)
# Observations	133	141	133	141	133	141	133	141	133	141
R^2	0.572	0.767	0.592	0.771	0.630	0.854	0.747	0.874	0.752	0.885

Standard errors are in parenthesis; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

effect is relatively strong in the case of strong energy poverty. Together, these results suggest that energy poverty incidence in terms of energy expenditure share is relatively low in the most densely populated rural areas and tends to increase in city size, except for the most dense cities.

Finally, in Table 8 we present the results of the same regression framework to explain energy poverty in terms of modern energy use (EP_{modern}). As expected, we find that this type of energy poverty is negatively correlated with per capita expenditure, education level and, albeit to a lesser extent, household size. Furthermore, we could not find a statically significant effect of the energy price for traditional biomass. However, we did find a statically significant negative effect of electrification on EP_{modern} in 2003, but this effect ceased to exist in 2015 –

presumably because the number of electrified district was by then so much higher that its distinctive effect had disappeared. As regards the spatial effects, it appears that EP_{modern} is lower in urban areas in general as well as in the Maputo area, albeit the effect is statistically not very robust except for the urban area effect in the strong poverty case (see Table C4 in the Annex). These findings are in line with what we expect: accessibility and affordability of modern energy is relatively good in urban areas, and even poor people in cities often consume at least some modern energy, hence especially strong modern energy poverty is statistically significantly lower in urban areas. Finally, we find evidence that EP_{modern} falls when population density increases, but rises again at higher levels of population density (regression model 4) – implying a U-shaped pattern. Again, this suggests that also in terms of modern energy use, energy poverty incidence tends to increase in city size.

The U-shaped and inverted N-shaped relationships between population density and (energy) poverty that emerge from our regression analysis are based on coefficients that reflect the predicted or fitted values of the relationship between population density and energy poverty, controlled for the potential influence of various other variables including household characteristics. Hence, the remaining question is as to what extend the predicted relationships between population density and energy poverty incidence hold for the actual data in our sample; in other words, at which point of the implied U-shaped and inverted N-shaped curve do the districts of Mozambique actually lie? To see this, we plot in Fig. 3 the predicted (energy) poverty levels for the entire distribution of districts in Mozambique in terms of their actual population density value and based on the regressions coefficients for population density from Tables 5–8 Depending on the poverty indicator, we use the preferred regression model version as indicated in the text above (either model 4 or 5). We do so, respectively, for 2003 and 2015 as well in terms of weak and strong energy poverty, with the actual population density values of each district plotted next to the horizontal axis as diamond shapes. Because we did not find a statistically significant independent relationship between population density and energy consumption poverty (EP_{GJ}), this relationship is not included in Fig. 3.

Fig. 3 shows that, after controlling for household and location characteristics (see Table 5), income poverty in Mozambique in 2015 (the orange line) is predicted to first fall with increasing population density, then rise, in order to fall again at the highest level of population density i.e. in the largest cities. In 2003 the latter effect was absent with the predicted income poverty incidence levels still increasing in the largest cities (the blue line). These patterns are similar for weak and strong income poverty, although more pronounced for the latter. These findings support the notion of urbanization of income poverty, although for 2015 this effect seems to be more prevalent for the mid-sized cities than for the largest city.

The predicted relationship between population density and energy poverty in terms of expenditure share (EP_{exp}) very much resembles that of income poverty in 2015: an inverted N-shaped pattern with EP_{exp} falling at low population density levels (i.e. in rural areas), rising at intermediate levels of population density, in order to fall again at the highest level of population density i.e. in the largest cities. This is different than in 2003 when EP_{exp} was actually declining exponentially in population density, with the lowest levels of EP_{exp} thus being achieved at the highest level of population density i.e. in the largest cities.

Finally, for the actual data in Mozambique we find an exponentially decreasing relationship between population density and energy poverty in terms of modern energy (EP_{modern}), conforming that this type of energy poverty evidently is most severe and persistent in rural areas. However, Fig. 3 also shows that in case of strong energy poverty and for the 2015 dataset, the predicted level of EP_{modern} just starts to increase at the highest population density level, i.e. in the capital city Maputo. The results thus suggest that if urban density continues to grow, EP_{modern} may start to rise again in the largest cities.

6. Conclusions

We studied the potential implications of urban growth in the global South on the evolution of energy poverty incidence. Remarkably enough, the relationship between urban population growth and energy poverty is yet a somewhat under-researched topic as these topics are mainly studied separately from each other in different strands of the literature. Our literature review demonstrated that from a theoretical perspective the relationship between increasing urban density and energy poverty incidence is ambiguous in advance. We therefore developed an empirical analysis to examine the evolution of (urban) energy poverty under influence of (urban) population growth, and we did so for the case of Mozambique – that exemplifies the future development of Africa's energy poverty situation in many respects. Specifically, we related for two years (2003 and 2015) the spatial variation in Mozambique's population density to the spatial variation in three commonly used indicators of energy poverty – low energy consumption quantities, a high energy expenditure share, and access to modern energy fuels – as well as a measure for income poverty for comparison. To this aim we made use of two newly constructed datasets with household level information that originate from national household welfare surveys as well as an own survey for Mozambique's capital city Maputo.

A descriptive analysis of our data led to the following observations. First, in contrast to rural areas, energy poverty is more severe than income poverty (except for strong energy expenditure poverty). Second, over time, and especially in urban areas, energy poverty, like income poverty, is generally decreasing in terms of consumption quantities and access to modern energy fuels. These results confirm previous findings in the energy poverty literature on relative high energy poverty incidence levels in rural areas. They also are line with the urban economics literature on agglomeration effects, in which densely populated areas are associated with rising incomes; hence, it is in those areas that we may expect increasing energy use and a gradual transition towards consuming modern fuels. but our analysis also revealed evidence of increasing energy poverty in urban areas, when measuring energy poverty in terms of energy expenditure shares. This increase in energy expenditure poverty is most pronounced in the capital city Maputo and concentrated among households that spend 10–20% of their income on energy. Third, and in addition, our data suggest that cities increasingly are home to a group of people that suffer from deep energy poverty, given the fact that strong energy consumption poverty is slightly more prevalent in urban than in rural areas. These latter results are in line with earlier observation of urbanization of poverty ([14]Ravallion 2002 [15],Ravaillon et al., 2007), but to the best of our knowledge have not yet been documented in the energy poverty literature. Finally, we showed that by largely ignoring transport energy use, the national welfare surveys are likely to overestimate energy consumption poverty with about 10% points while underestimating energy expenditure poverty with about 18% points. This finding underlines the importance of paying more attention to the role of urban transport consumption in measuring energy poverty incidence in the Global South, where especially urban households often spend a considerable share of their income on transport.

A multiple non-linear regression analysis provided evidence that population density directly impacts energy poverty incidence beyond its indirect impact through other location characteristics and spatial sorting of households. More specifically, we found that if population densities rise, energy poverty levels in terms of modern fuel share first fall and then increase – thus following a U-shaped pattern – while energy poverty levels in terms of energy expenditure, exhibit a similar pattern but then fall again at the highest level of population density – thus following an inverted N-shaped curve. To the best of our knowledge, these non-linear relationships between energy poverty and population density have not been documented before but are in line with the notion of urbanization of poverty in Africa. Economic development, which is first manifested in cities, typically comes with rising incomes, increasing energy use and a gradual transition towards consuming modern fuels, as is also manifest in our results. But, the non-linear nature of the link between energy poverty and population density also suggests that urbanization processes in poor countries like Mozambique may not lead to an unequivocal reduction of energy poverty rates, especially not as regards energy expenditure poverty.

Evidently, our analyses can be extended and improved in various respects. First, our conclusions ask for further research to see if the observed non-linear relationship between population density and energy poverty also occurs in other countries and in case of different and more sophisticated measures of energy poverty like, for example, the recently proposed alternative energy poverty framework proposed by Ref. [60]Pachauri and Rao (2020). Second, it must be seen whether our conclusions hold in case of better data, because measurement of urban (energy) poverty in poor countries like Mozambique is not straightforward. The national household welfare surveys in Mozambique, that serve as the basis for our first dataset, experienced difficulty capturing consumption in urban areas, notably for food ([69]Alfani et al., 2012 [70],Arndt et al., 2012 [29],Jones and Tvedten 2019). Thus far we have no evidence that this is also true for energy, but we cannot exclude the possibility that our results based on these data may produce biased estimates of urban energy poverty. Incorrect estimates of urban (energy) poverty in the global South is not uncommon (see e.g. Ref. [71]Chidebell-Emordi, 2015 [16],Lucci et al., 2018 [17],Mitlin and Satterthwaite 2013) and appears to be related to the complexities of capturing consumption in urban areas, the use of data collection methods and poverty measures that were designed more with rural households in mind (where poverty has traditionally been most widespread) and because the definition of basic (energy) needs varies by standard of living and regional locations.

Third, it is known that energy poverty is a problem that has a disproportionate effect on women and girls (see e.g. Ref. [72]Clancy et al., 2007, [73]Oparaocha and Dutta 2011). We did not analyze this gendered nature of energy poverty, but our results support the relevance of analyzing the role of gender in the urbanization of energy poverty: to what extent makes the urban context a difference in the gender-energy-poverty nexus? Finally, our results confirm the usefulness and need for more research into the interactions between urban form and energy use by households in the fast growing cities in the global South. After all, our research has shown that high levels of population density are not inevitably bringing prosperity and access to modern energy resources to all its inhabitants. A key question is therefore also how in fast-growing cities in poor countries, the design of policies can contribute to inclusive growth while speeding up the transition towards affordable access to modern energy fuels and a transport system that does not contribute to rising (energy) poverty levels. Often, energy (poverty) policies are considered in relative isolation by local governments; their design is frequently not integrated with, for example, transport policies and urban planning. Our research suggests that it is important to consider these issues in their interrelatedness, since they have a combined effect on the social welfare of the rapid growing group of urban people.

Credit author statement

Gilberto Mahumane: Conceptualization, Formal analysis, Software, Writing- Original draft preparation. **Peter Mulder:** Conceptualization, Methodology, Supervision, Writing- Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rser.2022.112089.

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